

## Subject Description Form

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| <b>Subject Code</b>                                   | EE4008 / EE4008A / EE4008B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Subject Title</b>                                  | Applied Digital Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Credit Value</b>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Level</b>                                          | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Pre-requisite/<br/>Co-requisite/<br/>Exclusion</b> | Pre-requisite for EE4008: EE3005<br>Pre-requisite for EE4008A: EE3005A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Objectives</b>                                     | <ol style="list-style-type: none"> <li>1. To facilitate a working knowledge of principles of reduced-order modelling, digital control algorithms, system identification, and adaptive control.</li> <li>2. To enable students designing industrial control systems for applications in different engineering areas.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Subject Intended Learning Outcomes</b>             | <p>Upon completion of the subject, students will be able to:</p> <ol style="list-style-type: none"> <li>a. Understand the concepts of reduced-order modelling, deadbeat control algorithm, system identification and adaptive control.</li> <li>b. Understand the notions of offline and online system identification.</li> <li>c. Design conventional and adaptive controllers based on user specifications.</li> <li>d. Use MATLAB simulation for digital system design and simulation.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b>      | <ol style="list-style-type: none"> <li>1. <b>Process control:</b> Process modelling, Performance Specification, Industrial controller, Ziegler &amp; Nichols tuning, Advanced process control, Reduced order modelling.</li> <li>2. <b>Elementary concept: digital control system:</b> Linear difference equations and the Z transform, Analog to digital and digital to analog converters, Zero order hold, Analysis of digital control, Real Implementation of digital control, Internal model control.</li> <li>3. <b>Digital PID control system design:</b> Discretization of PID control, Integral windup, Digital PID parameter tuning methods, 2DOF-PID</li> <li>4. <b>System identification:</b> Discrete-time and continuous-time systems, identification by correlation, principle of least squares, Recursive least squares.</li> <li>5. <b>Self-tuning control:</b> Introduction to adaptive control, Self-tuning controller.</li> </ol> <p><b>Laboratory Experiment:</b><br/>There will be two laboratory experiments on the topics of reduced order modeling, digital control design and system identification by least-squares technique.</p> <p><b>Case study:</b><br/>Individual assignment related to above methods. Students will write a report and present their finding to the class.</p> |

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| <b>Teaching/Learning Methodology</b>                                   | Lectures and tutorials are the primary means of conveying the basic concepts and theories. Experiments and case study are designed to supplement the lecturing materials. The students are encouraged to take extra readings and to look for relevant information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                   |          |          |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
|                                                                        | Teaching/Learning Methodology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                   | Outcomes |          |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
| a                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                   | b        | c        | d |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
|                                                                        | Lectures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                   | ✓        | ✓        | ✓ |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
|                                                                        | Tutorials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                   | ✓        | ✓        | ✓ |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
|                                                                        | Experiments and case study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                   |          |          | ✓ | ✓ |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
| <b>Assessment Methods in Alignment with Intended Learning Outcomes</b> | <table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="4">Intended subject learning outcomes to be assessed</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td></tr><tr><td>1. Examination</td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>2. Class test</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>3. Project report</td><td>10%</td><td></td><td></td><td></td><td></td></tr><tr><td>4. Case Study</td><td>10%</td><td></td><td></td><td></td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> <p>The outcomes on concepts, analysis and design are assessed by the usual means of examination and tests.</p> |                                                   |          |          |   |   | Specific assessment methods/tasks | % weighting | Intended subject learning outcomes to be assessed |  |  |  | a | b | c | d | 1. Examination | 60% | ✓ | ✓ | ✓ |  | 2. Class test | 20% | ✓ | ✓ | ✓ |  | 3. Project report | 10% |  |  |  |  | 4. Case Study | 10% |  |  |  |  | Total | 100% |  |  |  |  |
| Specific assessment methods/tasks                                      | % weighting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Intended subject learning outcomes to be assessed |          |          |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
|                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | a                                                 | b        | c        | d |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
| 1. Examination                                                         | 60%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ✓                                                 | ✓        | ✓        |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
| 2. Class test                                                          | 20%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | ✓                                                 | ✓        | ✓        |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
| 3. Project report                                                      | 10%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                   |          |          |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
| 4. Case Study                                                          | 10%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                   |          |          |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
| Total                                                                  | 100%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                   |          |          |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
| <b>Student Study Effort Expected</b>                                   | Class contact:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                   |          |          |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
|                                                                        | ▪ Lecture/Tutorial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                   |          | 33 Hrs.  |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
|                                                                        | ▪ Laboratory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                   |          | 6 Hrs.   |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
|                                                                        | Other student study effort:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                   |          |          |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
|                                                                        | ▪ Laboratory preparation/report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                   |          | 12 Hrs.  |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
|                                                                        | ▪ Case study preparation/report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                   |          | 14 Hrs.  |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
|                                                                        | ▪ Self-study                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                   |          | 40 Hrs.  |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
|                                                                        | Total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                   |          | 105 Hrs. |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |
| <b>Reading List and References</b>                                     | <b>Reference books:</b><br>1. Phillips, C.L., Nagle, H.T. and Chakraborty, A., Digital Control System Analysis & Design, Pearson, 2014<br>2. Franklin, G.F., J David Powell and Workman, M.L., Digital Control of Dynamic Systems. 3 <sup>rd</sup> Edition, Pearson Education, 2015                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                   |          |          |   |   |                                   |             |                                                   |  |  |  |   |   |   |   |                |     |   |   |   |  |               |     |   |   |   |  |                   |     |  |  |  |  |               |     |  |  |  |  |       |      |  |  |  |  |